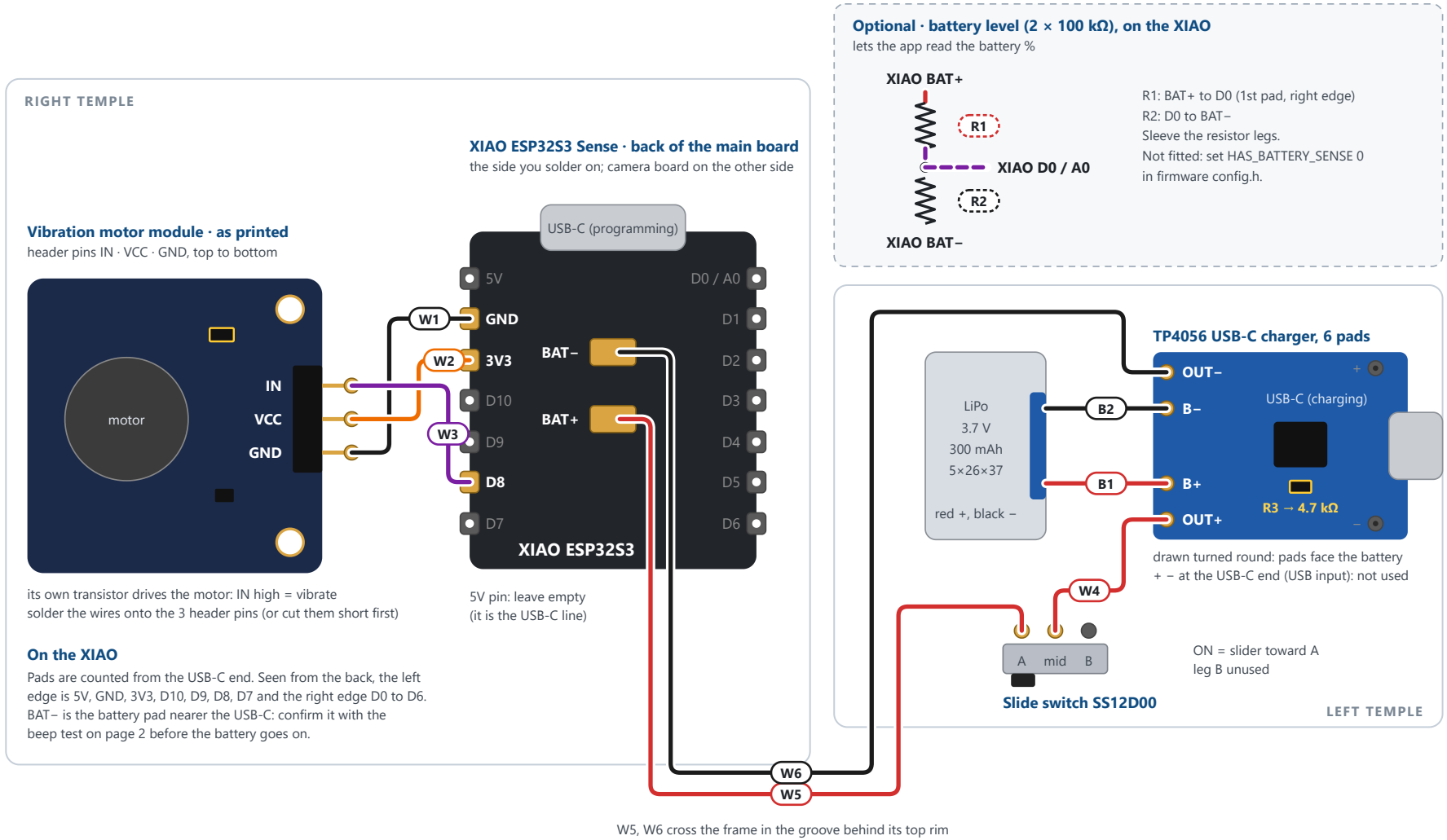
Baseera glasses — wiring diagram, stage 1

XIAO ESP32S3 Sense, vibration module, TP4056 USB-C charger, slide switch and 300 mAh LiPo, laid out as the glasses are seen from the front (right temple on the left). The XIAO is drawn from the back of its main board, where the wires are soldered: count its pads from the USB-C end. The module is drawn as printed, the charger turned round. Always solder by the printed pad name. Wire numbers match the wire list on page 2. Where two wires cross, the one drawn on top has a white edge: they are not joined.

battery + · GND / battery − · 3.3 V · motor control · optional



Wire list — Baseera glasses, stage 1

8 connections: W1–W3 in the right temple, B1, B2 and W4 in the left temple, W5 and W6 across the frame; R1 and R2 (grey) are optional. Every pad on the module, charger, switch and battery takes one wire; only the XIAO's pads take more: **BAT+ 2** (W5, R1) and **BAT− 2** (W6, R2) when R1/R2 are fitted.

#		Wire	From	To	Note
W1		black	module GND	XIAO GND (2nd pad from the USB-C)	right temple
W2		orange	module VCC	XIAO 3V3 (3rd pad)	the motor is rated 3 V
W3		purple	module IN	XIAO D8 (6th pad)	HIGH = vibrate (GPIO 7)
B1		battery red	LiPo red (+)	TP4056 B+	battery's own lead; solder this LAST (page 3)
B2		battery black	LiPo black (−)	TP4056 B−	battery's own lead; solder LAST
W4		red	TP4056 OUT+	switch MIDDLE leg	left temple
W5		red	switch leg A	XIAO BAT+ (battery pad farther from the USB-C)	across the frame; NOT the XIAO's 5V pin
W6		black	TP4056 OUT−	XIAO BAT− (battery pad nearer the USB-C)	across the frame; OUT−, not B−
R1		100 kΩ	XIAO BAT+	XIAO D0 / A0	optional battery level; resistor legs, sleeved
R2		100 kΩ	XIAO D0 / A0	XIAO BAT−	optional battery level

Wire and length

- 28–30 AWG silicone wire (BOM item 12), colours as above. If a colour is missing, use any colour and put a tape tag with the number on both ends.
- **W1–W3, right temple**: XIAO at the front of the right housing, module in the frame on its lid. About 8 cm each, so the lid can lie open beside the housing.
- **B1, B2, W4, left temple**: the charger lies at the back, the battery in front of it, the switch under the top wall at the front. B1 and B2 are the battery's own leads: shorten them to reach the charger's pads. W4 about 9 cm.
- **W5, W6, across the frame**: out of the left housing's wire slot, along the groove behind the frame's top rim, into the right housing's slot, to the back of the XIAO. W5 about 20 cm, W6 about 28 cm (it starts at the charger at the back), twisted loosely together.
- **On the XIAO's back**: strip 2 mm, tin, and lay each wire flat towards the back of the housing. Nothing may stand out past the board's edge by more than 2 mm: the channel beside the XIAO is 3.2 mm wide.

Checks before the battery goes on

1. **Which XIAO battery pad is which** (beep test): **XIAO GND** ↔ **BAT−** must beep; **GND** ↔ **BAT+** must not.
2. **XIAO 3V3** ↔ **GND** must not beep (a short chirp as the capacitors charge is normal; a steady beep is a short).
3. Switch OFF: **TP4056 OUT+** ↔ **XIAO BAT+** must not beep. Switch ON (toward A): it must beep.
4. **TP4056 OUT−** ↔ **XIAO GND** must beep.
5. Every row of the table: beep from the "From" end to the "To" end.
6. The battery itself: **red lead to black lead reads 3.0–4.2 V**. A minus sign means the colours are swapped: stop.
7. After the battery is on, switch OFF: **OUT+ to OUT−** reads the battery's voltage. If it reads 0 V, plug the charger's USB-C in once: that wakes the protection.

Never let the battery leads touch. Solder red to B+ first with the black lead taped. Nothing goes on the XIAO's 5V pin: it is wired to the XIAO's USB-C. One USB cable at a time while the switch is ON (the XIAO's USB-C also charges the battery through W5).

Build order and the charger fix

1 · Charger fix (before anything else)

This TP4056 board charges at **1 A**. The 300 mAh battery should get no more than about 0.3 A. Swap resistor **R3** (1.2 kΩ, marked **122**, next to the chip) for a **4.7 kΩ** resistor: about 250 mA.

1. Measure R3 first: about 1.2 kΩ (a little less is normal on the board). **If it reads something else, don't remove it**: send a photo of the board first.
2. Heat R3 and lift it off (a blob of solder across both ends helps).
3. Solder the 4.7 kΩ resistor's legs (bent short) onto the two R3 pads and trim them.
4. Measure across the pads again: about 4.7 kΩ.

2 · Order of work

1. Charger fix (above).
2. **Firmware** over the XIAO's USB-C: `firmware/baseera_glasses`, stage 1 in config.h (HAS_SPEAKER, HAS_DISTANCE, HAS_BUTTON 0; HAS_BATTERY_SENSE 0 unless R1/R2 are fitted).
3. **Bench test**: W1–W3 with the parts on the table, USB power only. At start-up the module gives one short buzz; then in the app join Wi-Fi Baseera-XXXX, connect, Test vibration.
4. Parts into the housings (right column). Solder W4 in the left temple.
5. Solder W5 and W6 to the XIAO's battery pads, thread them out through the right housing's slot, along the groove and in through the left housing's slot, then solder W5 to the switch's leg A and W6 to the charger's OUT−.
6. Checks on page 2, switch OFF, no battery.
7. **Battery last**: tape the black lead, solder red to B+, then untape and solder black to B−.
8. Switch ON without USB: one short buzz means it runs on the battery.

3 · Where each part sits (printed temples v4)

- **Right housing:** XIAO at the front, camera lens in the bezel, its USB-C under the top opening (programming).
- **Right lid:** vibration module in the printed frame, header edge forward, motor facing into the housing.
- **Left housing, back:** charger upside down near the floor, its USB-C in the groove of the back wall. The front edge rests on the rib; the rib on the lid holds it down.
- **Left housing, middle:** battery standing on the inner wall (foam tape), lead end towards the charger's pads.
- **Left housing, top front:** switch under the slot in the top wall (hot glue), right next to the wire slot.

4 · Good to know

- **Charging**: through the charger's USB-C in the back of the left temple, with the switch OFF or ON (red light = charging, blue/green = full). The switch only cuts the XIAO.
- The XIAO's 5V pin is its USB-C line, so the battery goes to **BAT+**, the XIAO's own battery input.
- OUT− (not B−) goes to the XIAO, so the charger's protection still cuts off a flat or shorted battery.
- D8 is shared with the camera board's microSD slot: leave the slot empty.
- The XIAO, its camera and Wi-Fi draw roughly 200–300 mA (estimate), so the 300 mAh battery lasts about an hour of use.
- When the phone link drops, the glasses buzz long-short-long by themselves.

Printed parts and their placement: `hardware/cad/PRINTING.md`.